Stephen James Kranz

EDITOR, ANIMATOR AND ENGINEER · BERKELEY, CALIFORNIA

stephenjameskranz@gmail.com
portfolio.calipercrayon.com

Editor and engineer. I've cut science and engineering videos with tens of millions of views, and I write, shoot and animate them too. I've spent as much time doing science and engineering as filming it.

EXPERIENCE

EngineerGuy · Producer and editor with Prof. Bill Hammack 2013 – 2015

- Co-wrote, directed, filmed, animated and edited explainer videos in Premiere Pro with tens of millions of views. The Ingenious Design of the Aluminum Beverage Can has 19 million.
- Built cutaway animations for what the camera couldn't see: the inside of a retractable pen, a music box, a NERF blaster.
- Photographed and designed a companion book to a video series on Albert Michelson's Harmonic Analyzer.

Autodesk · Digital storyteller, Ember 3D printer 2016

- Wrote, directed, filmed, illustrated and edited customer stories, feature explainers and how-tos within Autodesk brand standards. The Glidewell dental-crown story has 168,000 views.
- Researched print speed and quality; co-inventor on a patent for printing sharper vertical edges.

Origin Laboratories · Senior scientist, lab manager 2017 – 2020

- Managed an R&D 3D-printing lab of 40 printers and led a team of 3 scientists and technicians.
- Developed standardized procedures for evaluating the printability and optimizing the dimensional accuracy of photopolymer resins.
- Made safety and training videos for the lab.

Caliper & Crayon · Founder and designer 2020 – now

- House photographer for Church of Clown, San Francisco's clown theater: live comedy, low light, catching the joke on the right frame.
- Wrote a science highlight of an Advanced Light Source paper with five original diagrams.
- Designed the Magical Flag of Peace, made 671 hand-built variations, and built the web archive that catalogs them.

University of Illinois · Graduate researcher, Lewis lab 2010 – 2013

- Directed, filmed, edited and animated a video on the lab's silver ink pen for *Chemical & Engineering News* (about 250,000 views).
- Created photographs and figures for seven papers, including the inside front cover of *Advanced Materials*.

TOOLS

Premiere Pro, Cinema 4D, Illustrator, Photoshop, InDesign

SKILLS

Story editing · explainer and comedic pacing · motion graphics and cutaway animation · scriptwriting · shooting and lighting · hardware prototyping and 3D printing

SELECTED VIDEOS

The Ingenious Design of the Aluminum Beverage Can
EngineerGuy, 19 million views

Plastic Injection Molding
EngineerGuy, 4.8 million views

How a Wind Up Music Box Works
EngineerGuy, 2 million views

How to 3D print PFM dental crowns
Autodesk, 168,000 views

EDUCATION

M.S., Materials Science & Engineering
University of Illinois, 2013

B.S., Materials Science & Engineering
Arizona State University, 2010

PATENTS

Enhanced three-dimensional printing of vertical edges
US 10,780,641 B2, 2020

Multinozzle deposition system for direct write applications
WO 2013/006399 A2, 2013